

Contrasting Technologists’ and Activists’ Positions on Signing Avatars

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ABSTRACT

The research and development of signing avatars has become widely publicised with the promise to provide novel, efficient and accessible solutions to existing access issues deaf people face in their everyday lives. However, the deaf community is often lukewarm in their enthusiasm for signing avatars. There are deep structural issues present in this technology due to the systematic underestimation of the complexity entailed in sign languages, but also the quality of solutions and lack of direct participation of deaf communities. In my research, I explore the particular interrelationship between the deaf community and technical stakeholders regarding signing avatar technologies to analyse the gaps between community positions and technical realisation – particularly from a Human-Computer Interaction (HCI) perspective. My results indicate that signing avatar require more nuanced conceptualisations to properly identify adequate contexts for implementation and use that are acceptable to and accepted by the deaf community.

CCS CONCEPTS

• **Social and professional topics** → **People with disabilities**; • **Human-centered computing** → **Empirical studies in HCI**; Accessibility systems and tools; • **Information systems** → Language models.

KEYWORDS

Signing Avatars, Accessibility Technology, User-Centered Design

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1 RESEARCH PROBLEM & MOTIVATION

In recent years, researchers and developers have shown an increased interest in signing avatars¹ [31]. Subsequently, various parties such as government representatives and corporations promote such technologies as the latest barrier-free solution to their services,

¹I define a *signing avatar* here as a 3D representation of a human-alike character that signs content from written text in any sign language or International Sign.

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applications and products [30]. The use of sign language avatars is considered to be technologically advanced as a panacea for access issues of content that should be available in a sign language, particularly in the field of Sign Language Machine Translation (SLMT) and non-Western countries [18].

However, this perspective is generally not reflected within deaf communities, as shown by various recent statements of national and supranational deaf associations (e.g., [10, 13]). This critique not only addresses the systemic underestimation of the complexity of sign languages by technology developers, but also the resulting quality of offered solutions being perceived as sub-standard as well as a substantial lack of participation of deaf communities in the development of these technologies [8].

I draw on the perspectives from both, prior literature detailing the positions of deaf communities, and interviews with a selected group of deaf and hearing professionals concerned with the practical implementation of SLMT. In doing so, I aim to foster a mutual understanding of each viewpoint and derive potential strategies on how these could be more aligned where they differ. Through my analysis, I find that SLMT, while being discussed largely as either *helping* or *hindering* deaf people, requires a nuanced conceptualisation for all stakeholders to properly identify suitable contexts for implementation and use.

I now present some of the associated background information necessary to understand how deaf communities and technologists previously positioned themselves in the field. From this analysis, I derive the novelty of my approach before detailing my methods. After delineating the results of my research, I reflect on its contribution to the field of HCI particularly regarding the larger epistemological interest on how we may adequately involve marginalised people’s positions in developing technologies supposedly for them.

2 BACKGROUND & RELATED WORK

According to the World Federation of the Deaf (WFD), 70 million deaf people worldwide identify as members of the Deaf community. Their primary language is one of the over 200 globally available sign languages [24, 32]². Sign languages differ from spoken languages, as they consist of their own language system, grammar, etc. and generally understood as fully natural languages (cf. [16]). Sign languages do not have an agreed upon written form, which means that many deaf people experience spoken languages (in their written form) as a second language, which makes those less accessible to them [6].

Therefore, the idea of signing avatars responds to the additional current limitations inherent in video production with human

²I use the term deaf to include everyone with some degree of functional hearing loss and the term Deaf to additionally mark cultural belonging expressed through the preferred use of sign languages (akin to [22]).

signers, including production costs, inability for revisions in post-production and lack of anonymisation. Thus, especially animated signing avatars can be easily modifiable not just regarding their appearance, but also to customise the content in a simple and cost-effective way considering computational dynamic properties [20].

Even though signing avatars are considered highly advanced and barrier-free solutions by some technologists (e.g., [17]), there are critical statements, for example by the Austrian Deaf Association (Österreichischer Gehörlosenbund) and Austrian Sign Language Interpreters and Translators Association (Österreichischer Gebärdensprach-Dolmetscher:innen- und Übersetzer:innen-Verband) [13] which stress the concerns of signing avatars on the following key issues:

- (a) The inability to adapt to target groups with different background knowledge and experience with regards to a given subject matter
- (b) Weakening the visibility of deaf people in hearing-oriented publics
- (c) Lack of linguistic role models for deaf youths and hearing CODAs (Children of Deaf Adults), as well as
- (d) Penetrative cost comparisons without considerations of quality when aiming at replacing human interpreting or translating services.

On a supranational-level, the European Union of the Deaf (EUD) refers to the human right to sign language access for deaf people and states on a position paper that it is essential to develop sign language avatars considering a “constant and meaningful” involvement of deaf experts during the process of development and, among other things, to differentiate between *important (live emergency communications)* and *non-essential (weather forecasts)* information for the intended use cases of machine translation as this similarly pertains to the automatic translation between spoken languages [25]. On an international level, the WFD and the World Association of Sign Language Interpreters (WASLI) provided a statement indicating concerns about signing avatars and making a clear point for the argument that this technology cannot replace human sign language interpreters for various reasons including quality, flexibility and domain expertise [10]. Within HCI, there have been several nuanced analyses showing the complexity involved in catering to different groups of deaf people interacting with signing avatars [27] and the specific qualities they seek in animated vs. human signers [1]. Other work illustrates, by means of embodied critique, how some deaf people perceive sign language avatars to be unnatural [29], which tends to result in a detrimental loss of information that can be communicated through some avatars (compare the performance of the dominant signing avatar used in Austria, which yields a maximum of 52% rate of signing understandably without subtitles while at its worst performance this number drops to 15% [26]).

Despite the critical statements from deaf associations, Maartje de Meulder stresses the urgency of the ethical and responsible development of signing avatars [8]. Their work illustrates, among other aspects, the ideology from various signing avatars projects in terms of their assumptions about deaf people and sign languages. Non-signing representatives of these projects often claim that signing avatars can solve communication problems “for the deaf” quickly

(e.g., [15]). Thereby, comprehensive structural issues that deaf people face in their daily lives are simplified and reduced to quick technological fixes without critically considering them. To address these issues, it is essential to have a diverse team including deaf and hearing researchers or developers, to employ interdisciplinary methods, be aware of biases, and consider who the actually intended end-user is [8]. Kipp et al. highlight that the issue does not lie on the fact that “most researchers [of signing avatar projects] are not members of the Deaf community” but equally on the notion that “Deaf people as potential users have little or no knowledge about avatars” [20].

Recent research from the University of Vienna by Verena Krausnecker and Sandra Schügerl specifically addresses how a signing avatar could be designed. They provide a best-practice guide developers can consult when aiming to design such technologies for/with deaf communities [21]. However, this guide focuses largely on specific design characteristics, such as linguistic, translational, ethical aspects, situational appropriateness, social and aesthetic aspects. Their guidance does not include considerations regarding technical requirements or implementation factors to achieve these goals. My intended contribution lies in understanding and potentially closing this specific gap between qualitative criteria and technical requirements.

3 NOVELTY

As indicated in previous sections, prior research in the space of signing avatars tends to either consider a critical, social or a predominantly technical perspective. That means, that the works addressing the linguistic, translational socio and social, deaf community, end-user acceptance, ethical etc. perspectives, tend to do so exclusively, while simultaneously ignoring the constraining parameters involved in the technical background. These constraints are further difficult to assess due to their ever-changing and volatile nature. In addition to the conceptual perspectives on signing avatars, however, there are works that focus more on the technical realisation of signing avatars without fully addressing the social-linguistic backgrounds involved in holistically conceptualising sign language technologies. Thus, there is a gap of understanding, specifically regarding the interplay between these two entities: technical stakeholders of signing avatars and the acceptance of representatives of the deaf community as end-users but also as technological experts. For this reason, a bridge between the two parties is needed to capture their relative positions on the topic of signing avatars and to concretely dissect the specific implications different technical aspects can have on the potential of their use. Additionally, it was my interest to understand where the gap lies between technologists’ perspectives and activists’ critiques, how they are articulated respectively and how they could be reconciled productively. Hence, I explore precisely these perspectives with the aim to provide a consolidating, new angle on how technologists might develop artefacts for marginalised communities they are not themselves a member of.

4 APPROACH

The goal of my research is to examine different works dealing with signing avatars and to put a specific focus on the positions of

the deaf community, including cultural and linguistic perspectives. To then identify the different viewpoints and experiences of deaf and hearing individuals involved in the research, development and active commercialisation of signing avatars, I conducted four interviews with technical stakeholders, two with hearing and two with deaf individuals. All interviewees work in technical fields, albeit in different positions, e.g. managing director, assistant project leader or external expectation management. Furthermore, the projects they are involved in are more established (well-known) companies (i.e., *SignTime*³, *Kara Tech*⁴), national projects (*AVASAG*⁵) and supra-national projects established by the EU (*EASIER*⁶, *SignON*⁷). This broad set of different involvements in signing avatars allows me to draw on both, research-oriented and profit-oriented projects, which have different interests when it comes to the viability of avatars as well as their market feasibility. Specifically, I was interested in understanding how different stakeholders across the divide between science and economy differ in their assessments or perspectives. While researchers tend to approach their work through trial and error and are interested in exploring different tools and methods, for example, they may not always focus on the quality of solutions. On the other hand, companies tend to prioritise profit and economic survival, as this is critical for them to keep on existing, and tend to focus on fast and efficient solutions to be put to market quickly, sometimes leading to putting less emphasis on sufficient quality control.

I facilitated and conducted all interviews on Zoom and took between 40 and 60 minutes each. A (human) sign language interpreter was involved in every interview, but not always for the same purpose: In cases where I as a deaf researcher interviewed a deaf participant, the interview was conducted in German Sign Language (Deutsche Gebärdensprache; DGS) or International Sign (IS). So the sign language interpreter provided direct access to Dr. Katta Spiel (advisor) and aided the creation of a transcript. For the interview with hearing (non-signing) participants, the sign language interpreter additionally facilitated communication in English or German between me and the interviewee. The process of developing a transcript, which has been translated by a sign language interpreter and reviewed by me, presented its own challenges as sign languages do not have an associated agreed-upon writing system and must be translated into the written form of a spoken language for text-based analysis methods. Reviewing the transcript highlights that the translation may be mostly accurate, but does not always manage to capture the way I (or signing participants) intended to communicate or express myself (or themselves, respectively). Particularly subtle shifts and intentions get lost in translation from a spatial to a linear language. This highlights the challenge for deaf researchers in relinquishing control during interviews and in the transcription process, which builds the foundations for later analysis. The transcript further made me aware that due to the visibility of the dialogue, the problem may not always be in the interpretation; it is difficult to differentiate whether slight misunderstandings or unexpected answers were due to the 'difference' in modality (spoken vs

signed with the core meaning being the same) or the interviewee's interpretation of these questions regardless of modality.

Finally, the transcript was analysed drawing on the method of descriptive qualitative content analysis as delineated by Mayring [23]. The process starts with the identification of the source material (the interview transcripts), the analysis of the situated circumstances of its creation (cf. above), the formal characteristics of the material (cf. above), the intended direction of the inquiry as well as the specific question asked of the material (cf. Section 1). I then contrasted the individual interview transcripts and coded them for similarities and differences to identify and craft the themes presented in the following section. For this, I followed guidance on coding as developed by Boyatzis [5], first *sensing themes*, revisiting them together with the source material to *do so reliably* and *develop codes*. My presentation then focuses on the *interpretation of the information in the context of a theory or conceptual framework*. This combination of approaches to analysis allows me to flexibly draw on my core material as well as guiding sensitising concepts (i.e., [8]).

5 RESULTS

In this section, I present and discuss the five themes I identified through my interview analysis with the technical stakeholders and draw on select literature to critically contextualise these results.

Explicit vs. Implicit Ideologies: I noticed substantial differences between explicitly stated assumptions about sign languages and deaf communities and implicitly embedded ideologies in the technological artefacts. Understanding these distinctions is crucial when discussing the use of signing avatars (or SLMT more generally) [8]. Failing to connect a theoretical, critical understanding of deaf perspectives with matters of technical feasibility is a key problem affecting the overall product feasibility and acceptance of signing avatars. For example, one company (Interviewee A) uses their avatars (created through Motion Capture) to no small parts to translate children's books into sign languages. While this could be understood as appropriately addressing the critique of deaf communities regarding the use of signing avatars only in contexts of static texts, it entirely bypasses the issue regarding missing linguistic role models for children using and learning sign languages (cf. [13]). Lacking a deeper reflection on such issues becomes problematic, especially when hearing people without any substantial active experience with or use of sign languages in their daily lives, who do not fully grasp the complexities of Deaf culture, propagate the use of automated language technology, i.e. signing avatars, for everything, even in educational contexts (e.g., [28]). (Hearing) developers tend to use "the political language of the helping professions" [11], supposedly believing that technology is the answer to all problems faced by the deaf community, when in fact it may not be – as critically indicated by my deaf interviewees (B&D). Yet, both hearing interviewees were aware of the dominant critiques from the deaf community, despite their practices and technological approaches fully following through on implementing the resulting implications of that critique. For example, Interviewee C indicated that they used direct animation as their technical baseline due to the associated cost efficiency when having to implement small changes to a signed text. However, avatars created with this approach tend to be less comprehensible and are perceived as unnatural by deaf signers [19].

³<https://simax.media/>

⁴<https://www.kara.tech/>

⁵<https://avasag.de/>

⁶<https://www.project-easier.eu/>

⁷<https://signon-project.eu/>

Hence, I identify a mismatch between what is explicitly said in being aware of and responding to deaf communities' perspectives and the socio-technical practices embedded in the technological development of signing avatars.

Epistemic Exploitation: As described by Interviewee B, there can be an exploitative relationship between deaf and hearing colleagues in an inclusive team, particularly if the team only consists of one single deaf representative. Specifically, they shared their experience where they had the impression that deaf individuals may be expected to take on all responsibility for ensuring effective communication and adaptation within a mixed team, such as budgeting for interpreters, as this was not considered in project planning even though the project has the explicitly stated goal of accessibility and facilitation of communication. While the deaf expert was included to consult on matters of sign language specifics, they were additionally required to provide access to their non-signing, hearing colleagues.

It is important for the entire team to take responsibility for ensuring that all members have the resources they need to be successful, including the deaf. However, it is often the case that deaf team members are consistently dismissed or ignored as they must take on the additional work such as setting reminders, rescheduling, and organising sign language interpreters or repeatedly explaining perspectives from the deaf community. Hence, deaf experts are consulted as long as the information is provided in the modality of the dominant hearing stakeholders and the consultation provided has to be repeated and argued for in light of questioning judgements. This is an example for epistemic exploitation which "occurs when privileged persons compel marginalised persons to educate them about the nature of their oppression" [4].

In addition, it is important to understand that deaf people have an additional responsibility to their community, as the success or failure of a project involving sign avatars can affect the trust and perception of the deaf community. This underscores the importance of ensuring that deaf team members are not solely responsible for seeking feedback from their peers, as this puts them in the position of a double-bind where they have both responsibilities towards the team as well as towards their community and are singularly expected to resolve potential tensions. Within the commercial companies, this was slightly different in that deaf individuals were not necessarily acting as interfaces between hearing developers and deaf communities. However, deaf people were also not represented on the level of leadership, more as language or animation experts.

Similarly, when soliciting feedback, Interviewees B and D (both deaf) emphasised their felt responsibility towards the deaf community to give back to, i.e., in terms of educating their peers on technological feasibility or creating space for meetings that allowed for cultural exchange beyond the technological feedback sought of. They focused on qualitative methods that were oriented on exchange and dialogue, preferably in person. In contrast, Interviewees A and C indicated more formal, goal oriented methods where they solicited specific feedback in the form of surveys, questionnaires or dedicated focus groups. Here, the intent is to garner feedback unilaterally. Hence, even if the deaf community is involved there is nuance in how their involvement is facilitated or structured either as a dialogue or an extraction of feedback. It should be noted here

that the deaf community is not a monolith either. Different deaf people have different assessments, needs and desires when it comes to technology, including signing avatars.

Dealing with Small and Noisy Datasets. While, for spoken or written languages, particularly English, millions of datasets are used to train language models (cf. [3], also on the associated issues for the environment), for sign languages corpora of comparative size do usually not exist. However, more data is not always better when it comes to technological approaches for handling the problem of low amount and varying quality data. For sign languages, there is often a lack of data available for learning models, and even when data does exist, it may not be suited for the specific requirements of a given purpose. Most of the technologies used for the production of signing avatars are motion capture and animation or OpenPose. However, using these technologies can be challenging and resource-intensive, as they often require extensive post-processing and tweaking of settings. Identifying and recognising hands and fingers is relatively easy, but capturing nonverbal cues such as mouth gestures, facial expressions and constructed action can be more difficult, as the data for annotation is limited.

Annotating and coding tokens in collated data can additionally be time-consuming and resource-intensive, as it can take up to one hour of annotation for every one minute of video recording. If relying on small data sets too strongly, though, the reproduction of sign language through an avatar is severely limited. Additionally, a dictionary based approach assumes that sign languages can be neatly (and linearly) constructed through a set of lexemes when the choice of one sign over another is largely context dependant. Hence, all of my interviewees agreed that the use of artificial intelligence in this space, while potentially promising, currently is not sensible and manual animation or motion capture with growing dictionaries alongside will remain the technical state of the art for the foreseeable future.

Adequate involvement of deaf people is key. Being close to and working with the deaf community aids in better understanding the feedback, reactions, and needs of that community, particularly where needs differ between individuals. In general, though, deaf stakeholders have a better understanding of their needs, resulting in higher-quality feedback, more robust research and better products. However, it is important to remember that feedback is not always given and sometimes one must rely on their own assessment. Reasons for this may include survey fatigue, as deaf people receive too many surveys or requests to participate in interviews or focus groups [7] where they feel their representation is limited. Furthermore, some members of the deaf community in Austria have had negative experiences and may not want to give feedback [21].

As Interviewee D clarified, a successful involvement means that the deaf community prefers direct communication, so it is important to show&tell them prototypes and collect direct feedback. According to D, there is a myth that deaf people are technophobic, but it is important to know that they are not against technology, but they want to be involved in its development and not just be passive recipients. In addition, the technology should be useful and not like signing gloves that are not useful for them, as hearing researchers may develop things like signing gloves but deaf people do not want them [12]. Therefore, continuous feedback sessions throughout

the project, not just at the end, are crucial to not just garner any feedback but useful and productive feedback. It is further important to consider diversity within the deaf community, such as Black, Brown, Asian Deaf, Deaf+, and DeafBlind individuals [2].

Finally, there may be an information gap if deaf people do not have access to information about the various careers available to them. This is largely due to the lack of access to academic institutions in many countries, limited information about career paths, and the dependency on interpreters [9], who may not always be available or have the required background knowledge. Hence, multiple layers of oppression and differing experiences need to be carefully considered when planning the involvement of deaf people – particularly when they are expected to have both technical and linguistic expertise.

Acceptance needs time. As technology designed specifically for the deaf community advances, its development requires collaboration, educational outreach, deaf community involvement, and time to promote acceptance, both of the technology as such but also a consideration of deaf people's needs by hearing people. To understand the following statements, it is important to know that the deaf community is overburdened with the barriers they face in daily life.

For example, Interviewee D mentioned a story where one 60-year-old person compared the development of avatars to the advent of Video Relay Service (VRS) interpreting: Some deaf people initially hated it entirely and preferred human interpreters on site. However, acceptance has changed over time so that it is now in commonplace use – even if not considered to be on par with in person human interpretation.

However, in terms of signing avatars, there are additional concerns that governments might only fund presumably cheap machine translations and not human interpreters, making the deaf community not only reliant on interpreting services [9], but additionally on ones developed largely by groups of people without sufficient signing capabilities.

Additionally, some members of the deaf community have high expectations for signing avatars, hoping for a magical solution that these can simply replace the current problems with sign language interpreters (quality, availability, payer issue, etc.). This is why careful expectation management is so important. There needs to be a balance between acknowledging how things *should* work as well as the limits on how they *can* work. Hence, due to the persistent experiences of exclusion made in everyday lives, many deaf people yearn for a technical solution that would provide them access. It is especially important here not to oversell the capacities of signing avatars and create false hopes about the functionality and feasibility of the technology.

Conclusion for Signing Avatars. In summary, researchers and developers in the space of signing avatars need to recognise the substantial contrast in purposes different projects identify in the use of this technology. This contrast represents a potential conflict: While companies are interested in selling signing avatars as a product, research projects are more about experimenting methods and techniques (partly considering AI technologies).

Through my interviews, I noticed that deaf people are open to new technologies in principle, but also have expectations regarding

their meaningfulness and applicability in daily life as well as direct participation in development. This could be seen as an advantage, especially in the feedback process for quality assurance of the design. Not only is participation in the feedback process relevant, but also in the process as a whole. This means that more deaf people need to be encouraged in and provided access to the academic or technical fields to make deaf technology experts more widely available, as technology experts so far tend to predominantly be hearing.

In addition, the use of signing avatars needs to be chosen with special consideration, and critical, low-threshold information needs to be distinguished to avoid potential negative effects. On the one hand, signing avatars can partially provide a solution to existing problems, such as the availability of interpreters in emergency situations (until a human one is available to ensure information has been transmitted correctly, in the appropriate style and comprehensiveness), better accessibility of media for simple texts, but they are not suitable for vital, sensitive or intimate information (e.g., psychotherapy), which of course means limiting the areas of application – although these areas are similarly complicated with the attendance of human interpreters [9].

My results then indicate that signing avatars do have the potential to open up new opportunities in ad-hoc translation, the accessibility of some media content or even anonymisation of signed content; they could *help*. However, given the dominance of hearing, non-signing people in the field, the baseline for bringing products to market becomes fairly low leading to issues with avatars being incomprehensible. This, in return, leads to poor acceptance with the deaf community feeling steamrolled, effectively *hindering* not just them in contributing to technologies for them, but also the interests of those creating signing avatars. Hence, HCI requires more discussions and considerations of sign languages as part of different cultures and the (technological) limits of signing avatars.

6 CONTRIBUTIONS TO HCI

As a deaf researcher, I contributed an interview study with hearing and deaf stakeholders in different projects concerned with researching and developing signing avatars. My research shows that the singular involvement of individual deaf people in the project development of signing avatars is not sufficient to bring the deaf community on board. However, I could also identify issues in the availability of deaf technical experts based on the pervasive educational exclusions experienced throughout generations (e.g., in Scotland [14]). As Interviewee B describes the dilemma: “Highly qualified deaf people (in the technical field) either have to choose a hearing technical team to make use of their skills, but face limitations in communication, or choose a deaf expert team for sign language and face limitations in the use of their technical skills, whereas communication is guaranteed.” My analysis illustrates vividly how we need to redistribute the responsibilities on how access is provided and participation is structured within technology concerned teams. Without questioning the access related issues deaf people experience in their daily lives more generally, including their socially constructed components and power dimensions, no technological artefact will solve these issues.

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